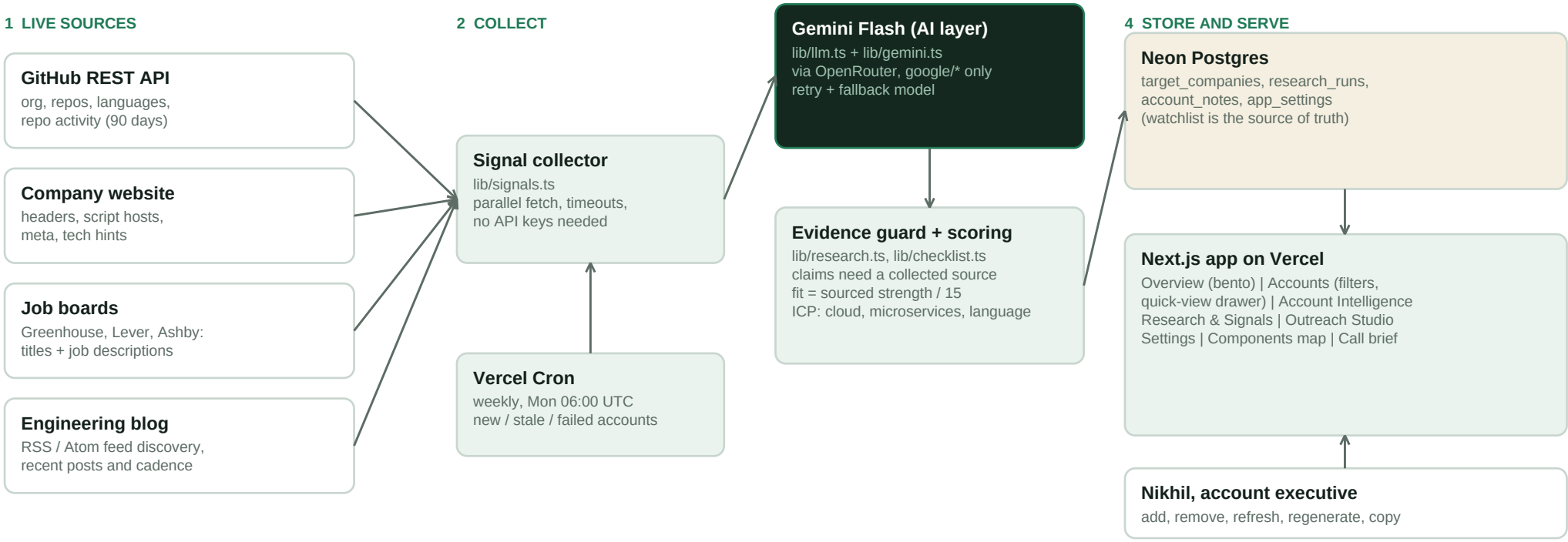
Stack Scout: Components Map

Live app: https://tech-stack-scout-topaz.vercel.app | Repository: github.com/manasvinipant-cloud9/tech-stack-scout

Public engineering signals > Gemini Flash > evidence-checked intelligence in Neon Postgres > Next.js app, refreshed weekly on Vercel



5 HOW THE APP CALLS THE PIPELINE

Add, refresh or bulk-refresh	**POST /api/companies/[id]/refresh**	Same path as the weekly job: collector > Gemini > guard > Neon, plus a run record
Regenerate the opening line	**POST /api/companies/[id]/opening-line**	Gemini Flash, grounded in the stored signals; line + its source saved to Neon
Settings: ICP targets, stale days, weekly limit	**GET, PUT /api/settings**	Saved in Neon (app_settings); read by the ICP checklist, stale rules and the weekly job

External source or person | Application code | AI (Gemini Flash only) | Stored data

Manual refresh and opening-line regeneration (from the app) call the same collector and AI layer as the weekly job.

Components

Every component, what it does, and where it lives in the repository.

Component	Responsibility	Technology	Code
Signal collector	Pulls public engineering signals from four live sources in parallel	fetch, Next.js route handlers	`lib/signals.ts`
Job-description reader	Reads up to 15 engineering posts; counts tool and microservices mentions	Greenhouse / Lever / Ashby public APIs	`lib/signals.ts`
Engineering-blog reader	Discovers a feed on common blog locations; reads recent posts and cadence	RSS / Atom parsing	`lib/signals.ts`
AI layer	One gateway for every model call: retries, fallback model, structured JSON. Only Gemini Flash models are called	OpenRouter (google/gemini-3.8-flash, fallback 3.7-flash, Google-only providers) or Gemini API directly	`lib/llm.ts`
Gemini analyst	Classifies the stack (confirmed or inferred, with source), judges architecture, writes fit reasons, open questions, the two-paragraph summary and the first line	Gemini Flash	`lib/gemini.ts`
Evidence guard	A claim citing a source that was not collected is downgraded to inferred / no source	TypeScript	`lib/gemini.ts`
Fit score	Sum of sourced fit-reason strength (1 to 3) divided by 15; unsourced reasons score 0	TypeScript	`lib/research.ts`
ICP checklist	Cloud, microservices and language checks against configurable targets, each with a verdict and evidence	TypeScript	`lib/checklist.ts`
Coverage model	Per account, six signals in five states: evidence found, partial, no evidence found, not researched, research failed	TypeScript	`lib/coverage.ts`
Action queue and insights	Prioritised next actions; changes and new signals computed from stored records	TypeScript	`lib/actions.ts`, `lib/insights.ts`
Opening-line generator	Regenerates a first line grounded in one verified signal, stored with its source	Gemini Flash	`lib/outreach.ts`
Watchlist store	Companies, cached intelligence, signals, notes, run history, settings	Neon serverless Postgres	`lib/db.ts`
Weekly refresh	Re-researches new, stale or failed accounts up to a saved per-run limit; secret-protected	Vercel Cron	`app/api/cron/refresh`, `vercel.json`
Application	Sidebar shell, command palette, bento Overview, filterable Accounts with quick-view drawer, Account Intelligence, Research & Signals, Outreach Studio, Settings	Next.js App Router, Tailwind v4	`app/(app)`, `components/`

Data model, pages and API

Neon Postgres tables

Table	Holds
target_companies	The watchlist: name, domain, GitHub org, status, fit score, stored analysis and signals (JSON), summary, opening line
research_runs	One row per refresh: status, fit score, previous fit, what changed, error
account_notes	The user's notes per company, with tags
app_settings	Saved ICP targets, stale threshold, weekly per-run limit

Pages

Page	Purpose
/ Overview	Priorities, next actions, fit distribution, signal coverage, cloud and language mix, fit vs hiring, changes, research timeline
/accounts	Watchlist: search, saved views, filters, sorting, bulk actions, add and remove, quick-view drawer
/accounts/[id]	Account Intelligence: summary, ICP fit and evidence, technology map, activity, opening line, methodology
/research	Coverage matrix, freshness, evidence gaps, refresh queue, run log
/outreach	Choose a company, inspect observations, generate and copy an opening line
/settings	Edit ICP and research preferences; read-only integration status
/brief/[id]	Printable one-page call brief
/components-map	This map, live in the app

API routes

Route	Purpose
POST /api/companies	Add a company
DELETE /api/companies/[id]	Remove a company
POST /api/companies/[id]/refresh	Re-research one company
POST /api/companies/[id]/opening-line	Regenerate the opening line
GET /api/companies/[id]/preview	Quick-view data for the Accounts drawer
GET, POST /api/companies/[id]/notes	Notes
GET, PUT /api/settings	ICP and research preferences
GET /api/cron/refresh	Weekly job (requires CRON_SECRET)

How the brief's requirements are met

Requirement	Where
Live external data	GitHub, website, job boards, engineering blog
All AI on Gemini Flash	lib/llm.ts: only Gemini Flash models are called
Languages, cloud tooling, team activity per company	Account Intelligence: technology map and activity
Two-paragraph summary and a non-template first line	lib/gemini.ts and lib/outreach.ts
Relevance in cloud, microservices and language	ICP checklist, shown on every account
Refreshed weekly	Vercel Cron, Mondays 06:00 UTC
Add and remove companies without touching code	Accounts page dialogs, or the database directly
Deployed on Vercel with a public URL	https://tech-stack-scout-topaz.vercel.app